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REVIEW OF TEXT SUMMARIZER TECHNIQUES

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Abstract : Text Summarization is process in which important information is extracted or collected from large amount of textual data. In this paper we are going to review and compare various approaches of text summarization including extractive methods like Frequency-Based Methods, Graph-Based Approaches, Machine Learning-Based Approaches and abstractive methods like Seq2Seq with LSTM/GRU, Seq2Seq with Attention Mechanism, Pointer-Generator Networks, Transformers and Pre-trained Models, Reinforcement Learning-Based Models. Our primary objective is to gain more insight about these methods like their strengths, limitations and applicability.

Index Terms - Text summarization, Extractive summarization, abstractive summarization, TF-IDF, TextRank Algorithm, LSTM, seq2seq Pointer-Generator Networks, Context Vector, Transformers.

I . INTRODUCTION

Text Summarization is a process of distilling the most important information from a source. Introduction Text summarization plays a crucial role in humans' daily lives as there is an immense amount of textual data being created every day. Due to the fact most of the information around us, such as news, emails, and articles, etc., is mostly more than what we can handle in a single sitting, the need for summarization has risen, and it is helpful to ease the decision-making process by reducing information overload and making it efficient.

There are two major classifications of text summarization: extractive and abstractive. Extractive summarization is the extraction of important sentences from the source itself, and abstractive summarization synthesizes a short summary through comprehension that expresses critical content. The goal of this paper is to present one-line descriptions for commonly used extractive and abstractive methods, along with their implementations, benefits, and drawbacks.

II. RESEARCH METHODOLOGY

2.1 EXTRACTIVE SUMMARIZATION APPROACHES

Extractive summarization selects key sentences directly from the source without rephrasing, making it computationally simpler and often more grammatically consistent.

2.2 Frequency-based approach

This models summaries text based on frequency of word. His algorithm works by removing stopwords from the text and then tokenizing it and ranking sentences based on repeated words is used for generating summary. Techniques such as Term Frequency- Inverse Document Frequency (TF-IDF), Natural Language Processing (NLP) using NLTK are used. It identifies high-frequency words to form summaries but may miss important low-frequency.

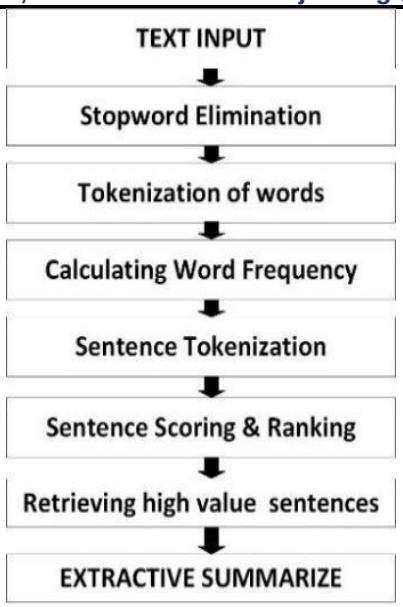

Fig1 Extractive summarizer (Frequency-based)

2.3 GRAPH-BASED APPROACHES

In graph-based models [1], sentences are represented as nodes while edges represent degree of their similarity. A well-known example is TextRank algorithm. In TextRank, sentences with more connections to other sentences are considered as more important for the summary. These sentences are selected to create a summary.

2.4 Methods For Abstractive Summarization

Abstractive summarization means it summarizes the text in few words with thought of key points and gain its meaning.

2.4.1 Seq2Seq with LSTM/GRU

The Seq2Seq architecture [2][7] consists of an encoder-decoder mechanism, which takes the input string and converts it to a summary. So, how does it work… the encoder processes the input text and converts the information to a single vector, called a context vector. Vector contains the meaning of whole text. The decoder takes this context vector and creates a summary, one word at a time. However, the model has issues dealing with longer documents because context vector is fixed length.

2.4.2 Seq2Seq with LSTM and attention mechanism

As Seq2Seq is generating each word of the summary, attention [3] mechanism provides a useful feature for the model to focus on the most relevant part of the input text. Instead of a single context vector, attention allows the model to "pay attention" to one or more words/phrases from the original text at any step of decoding. Such dynamism is what makes summaries shorter, intermediate, and longer documents more accurate and coherent as the model can change its focus based on whatever is most relevant at a given point when summarizing content.

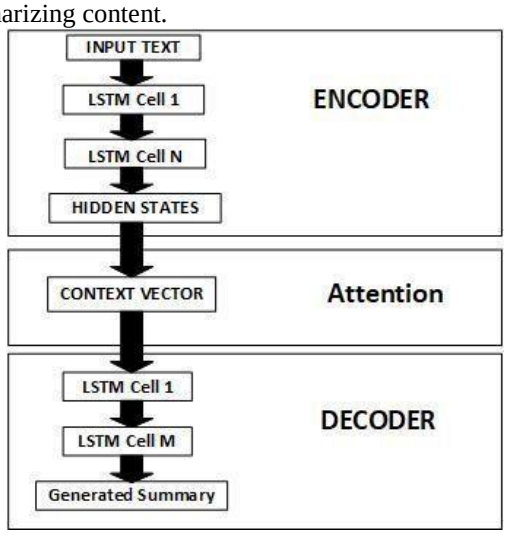

Fig 3: SeqtoSeq with LSTM and Attention mechanisms

3. Pointer-Generator Networks

Pointer-Generator [4] is a hybrid between extraction and abstraction, allowing the model to copy words from the source article or to generate them. Leveraging this approach allows processing of out-of-vocabulary words and to produce phonetically cohering summaries with paraphrased root word terms.

4. Transformers and Pre-trained Models

The ability to perform self-attention on large corpora with extensive pre-training using transformer-based models like BERTSUM and Pegasus [5]. They perform astoundingly well at generating summaries especially for abstractive summarization — with little to no additional tuning. More importantly, the self-attention mechanism allows them to attend to different parts of the input text with varying degrees of importance and thus easily capture complex relationships across sentences.

5. Deep Reinforcement Learning-Based Models

Evaluating methods are divided into traditional and extensive approaches based on the style of producing sentences using Natural Language Processing models. Traditional summarization methods [5] consider summarization as a goal-driven process, where the deep reinforcement learning (RL) model was used to train by maximizing evaluation metrics such as ROUGE. Here, the model builds sentences on the fly and gets reward for each step depending how similar this step is to an ideal summary. Over time the model continues to adjust its output, coming back with a more succinct summary which is topical. While these models based on RL are effective, they are computationally intensive since they need to be trained from scratch.

III. RESULTS AND DISCUSSION

3.1. Comparison

Comparison between Extractive and abstractive summarization-

Feature	**Extractive Summarization**	**Abstractive Summarization**
Core Technique	Sentence Selection	Text Generation
Computational Cost	Lower	Higher
Grammatical Accuracy	High (preserves original text)	Lower (can introduce errors)
Flexibility	Lower (limited to source text)	Higher (can generate new text)

Comparison between Abstractive approaches of summarization

Model	**Core Idea**	**Strengths**	**Weaknesses**
Seq2Seq with LSTM/GRU	Encodes input into a fixed-length vector, decodes into a summary	Simple, handles variable-length sequences	Suffers from vanishing gradient, struggles with long sequences
Seq2Seq with Attention Mechanism	Enhances Seq2Seq by focusing on relevant input parts	Improved performance, better handling of long sequences	Relies on RNNs, computationally expensive
Pointer-Generator Networks	Combines extraction and abstraction	Accurate and fluent summaries, handles out-of-vocabulary words	Complex architecture, requires tuning
Transformers and Pre-trained Models	Leverages self-attention for long-range dependencies	State-of-the-art performance, handles long sequences, generates coherent summaries	Requires significant resources, less interpretable
Deep Reinforcement Learning-Based Models	Trains model to maximize reward function	Learns complex strategies, adapts to tasks	Requires significant resources and data, sensitive to hyperparameters

The paper is organized as follows, here we have reviewed with the available works in text abstraction summarization. We have explored different approaches, i.e., Seq2Seq models, attention networks and transformer architecture. Upon looking deeply, we found that Seq2Seq with Attention Mechanism is highly potent and efficient model to yield compact Bullet Point Summary.

The model is vastly improved with the introduction of attention mechanisms to capture long-range dependencies and emphasize relevant text in the input. It results in summaries that are more coherent and relevant to the context. Although transformer-based models have been pulling out better results through weeks, but they require a huge computational power that cannot be applicable in every situation.

Finally, Seq2Seq with Attention Mechanism provides a good trade-off for both performance and computation. Through optimal architecture design, hyperparameter selection and extraction of the best quality data sets, it can be used to lay effective foundations for achieving complex summarization systems.

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